KneeMind v12

How the approach works: preprocessing, labels, model, training, inference

RSNA knee MRI competition (Kaggle) | companion to kneemind-v12.ipynb

This document explains what the notebook does and why, in enough detail to follow the code or to rebuild the idea elsewhere. Every number comes from the notebook, its config, or the saved training logs of the v12 run. Where something was not measured, the text says so.

At a glance

Item	Value
Task	Score each knee MRI study for 12 findings: ACL, MCL, medial meniscus, lateral meniscus, medial OA, lateral OA, patellofemoral OA, effusion, synovitis, Baker's cyst, contusion, fracture. Metric: AUC averaged over findings.
Training data	4,407 studies: 4,349 with report-derived (weak) labels and 58 with expert (gold) labels. The 58 are never trained on.
Input per study	Up to 5 series (3 sagittal/coronal/axial fluid-sensitive and 2 non-fluid), each resampled to 336 x 336 px at 0.457 mm (153.6 mm field of view), 12 slices per series over the central 15-85 %.
Model	ConvNeXt-nano 2.5D encoder, 2-layer study transformer, one attention-pool query per finding. About 19 million parameters per member (75.8 MB in fp32).
Ensemble	4 members: 2 trained in v12 (seeds 142, 143) + 2 from v11, plus 2 slice-set passes (TTA).
Training	24 epochs, AdamW, EMA weights, BCE with positive weights. 239 and 263 minutes for the two members, trained side by side on two T4 GPUs.
Scores	v11 hidden set 0.908 (v7: 0.897). v12 gold-58 AUC of the members: 0.881 and 0.899. v12 ensemble hidden score: not recorded.

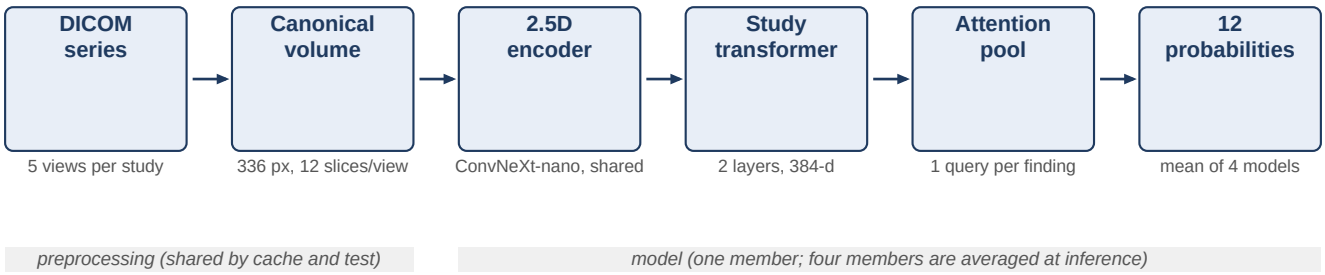


Figure 1. End-to-end flow. Preprocessing is identical at training and test time.

1. Problem and design logic

Each study contains several MRI series (different planes and contrasts), each a stack of DICOM slices. The competition asks for a probability for every one of the 12 findings, and ranks submissions by AUC averaged over findings. AUC depends only on the ordering of studies within each finding, not on calibration.

The training set has an unusual shape. Only **58 studies** carry expert labels (gold). For the other **4,349**, the only supervision is the free-text **radiology report**, which has to be turned into labels automatically. That has three consequences, and the whole design follows from them:

- **Labels are the bottleneck.** A report can be wrong, vague, or in another language. Better label extraction helps more than a bigger image model, so a large part of the notebook (section 3) is about labels.
- **The 58 gold studies are too few to tune on.** With about 10 positives per finding, the standard error of the gold AUC is roughly 0.02. They are used only as an honest validation set and to weigh label sources, never for training, and the last-epoch weights are kept instead of picking an epoch by gold score.

- **The same code must run on train and test.** Preprocessing is one function used for caching and for the hidden-set run, so the two cannot drift apart.

2. Preprocessing: from DICOM series to a fixed-size volume

Every study is converted by *process_study* into a uint8 array of shape [5 views, 12 slices, 336, 336], a mask saying which views exist, and a 3-way side tag. The result is cached once as a compressed .npz per study (about 4,407 files) so training never touches DICOM again.

2.1 Choosing the series

A study may have many series. For each of the 5 view slots (sagittal fluid-sensitive, coronal fluid-sensitive, axial fluid-sensitive, sagittal non-fluid, coronal non-fluid) the code picks one series: first an exact match on anatomical plane and fluid-sensitivity flag, otherwise the same plane with the other flag. A series is never used for two slots. If a slot has no series, it is simply masked out.

2.2 Building the volume

Step	What happens
Slice order	Frames are sorted along the true slice normal (cross product of the DICOM row and column direction vectors), not by file name. Fallback: InstanceNumber. Enhanced multi-frame DICOMs are handled.
Pixel values	Rescale slope and intercept applied; MONOCHROME1 images inverted.
Spacing	Each frame is resampled to 0.457 mm per pixel (area interpolation when shrinking, linear when enlarging), then centre-cropped or zero-padded to 336 x 336. 336 x 0.457 mm = 153.6 mm field of view for every study.
Slice sampling	12 evenly spaced slices over the central 15-85 % of the stack. The outer 15 % at each end is mostly empty or off the joint.
Intensity window	Per series: the 1st to 99.5th percentile of non-zero voxels is mapped to 0-255.
Orientation	Axes are flipped so row and column directions point toward the patient's positive axes. Left knees are **not** mirrored in v12 (see 2.3).
Failures	A slice that cannot be decoded is zero-filled; if more than 40 % of the sampled slices in a series fail, the whole series is treated as missing. A black volume is never passed on as a real view.

2.3 Side tag instead of mirroring

The notebook reads laterality from the DICOM Laterality tags, or from the series, study and protocol descriptions (several languages), and stores it as a 3-way one-hot (left, right, unknown). The side tag exists on only about half of the series, and a public A/B study reported that mirroring left knees to one orientation on such an unreliable tag hurt clearly (-0.10 macro AUC after 3 epochs). So v12 keeps the raw orientation and feeds the side tag to the model as an input; the model learns what to do with it.

3. Turning reports into training labels

Four report-derived label tables and one rule-based extractor are combined into a single soft label (a value from 0 to 1) per study and finding.

3.1 Rule-based extractor (regex labeler)

The notebook has its own multilingual regex labeler (kept from v5). It lowercases and strips accents, splits the report into sentences, and for each finding looks for a structure term (for example ACL or medial meniscus) together with a finding term (tear, rupture, effusion, ...). The clause around the match is then checked:

- a negation before or after the term (for example "no", "without", "negative for", and their translations) discards the match;
- severity terms raise the score to 1.0; intermediate terms set a middle value; exclusion terms (such as mild or old changes) set 0;
- otherwise the finding gets a base score. The score of a finding is the maximum over all sentences.

Finding	Base score	Finding	Base score
ACL	1.0	Effusion	0.6
MCL	1.0 (injury wording: 0.3)	Synovitis	1.0
Medial / lateral meniscus	0.6	Baker's cyst	0.6
Medial / lateral / PF OA	0.3	Contusion	0.7
		Fracture	1.0

Table 1. Base scores of the regex labeler before the severity and negation rules.

3.2 Label fusion weighted by agreement with the gold studies

Any attached CSV that has a study-id column and at least 8 of the 12 findings is picked up automatically as another label source. Here that gave four LLM-extracted tables from public datasets (by stevenleehans: three versions; by pilkwang: one). For every source and every finding, the AUC against the 58 gold labels is measured, and the source gets the weight **max(AUC - 0.5, 0.02)^2** for that finding only. The fused label is the weighted mean over the sources that have a value. A source that is poor on synovitis therefore has little say on synovitis, but keeps its say on ACL.

Source (AUC vs gold)	ACL	MCL	Med. men.	Lat. men.	Med. OA	Lat. OA	PF OA	Eff.	Syn.	Bkr	Con.	Fx	Mean
regex v5 (own)	.950	.981	.820	.786	.826	.781	.716	.835	.639	.920	.822	.786	.822
LLM full	.993	.964	.954	.862	.931	.830	.901	.853	.678	.946	.831	.793	.878
LLM v2	.993	.964	.954	.862	.931	.830	.901	.853	.790	.946	.831	.793	.887
LLM v4 blend	.987	.968	.948	.879	.932	.833	.902	.877	.790	.944	.860	.793	.893
pilkwang v2	.997	.976	.943	.841	.908	.789	.891	.830	.694	.932	.768	.871	.870
Fused	**.994**	**.982**	**.947**	**.873**	**.931**	**.815**	**.907**	**.858**	**.784**	**.947**	**.838**	**.812**	**.891**

Table 2. AUC of each label source against the 58 gold studies (from label_sources_vs_gold.csv of the v12 run). Fusion weights are fit on these same 58 studies, so the fused row is mildly optimistic.

3.3 Leak guard

Some public label tables copy the gold labels. Such a table would score a near-perfect AUC on the gold studies and take almost all the weight, which would then make the gold validation meaningless. The notebook therefore checks, for every table, how many of the gold cells it reproduces exactly. At 90 % or more it is flagged as a leak and gets the average weight of the clean sources instead (it may still label the other 4,349 studies well). If a leaky source is present, the fused gold AUC is printed as not trustworthy. In the v12 run no source was flagged.

4. Model architecture

One study goes in, 12 probabilities come out. The same ConvNeXt encoder looks at every sampled slice of every view; a small transformer then lets the evidence from different views talk to each other; and one attention query per finding collects the evidence for that finding.

Why these parts (data flow in Figure 2)

- **2.5D input.** Each slice enters the 2D encoder as three channels (previous, current, next cached slice), giving the network a little through-plane context without a 3D network. Because 12 slices are spread over 70 % of the stack, neighbours are the neighbouring *cached* slices, not necessarily adjacent anatomical slices.
- **View and position embeddings.** Every token knows which view it comes from and where in the stack it sits (continuous position from 0 to 1).
- **Study transformer.** Self-attention over all tokens (30 at test time: 5 views x 6 slices) lets sagittal and coronal evidence about the same structure interact. The side tag is added to every token.
- **One attention query per finding.** The ACL query can concentrate on two or three sagittal slices, while the effusion query can spread across the suprapatellar pouch. The pooling head itself is light: 12 query vectors and 12 weight vectors.

- **Masks.** Missing views are masked in both the transformer and the pooling, so a study with only 3 series is handled without special cases. A study with no views attends to zeros instead of producing NaN.
- **Precision.** The encoder and transformer run in fp16 autocast; pooling and the final head run in fp32.

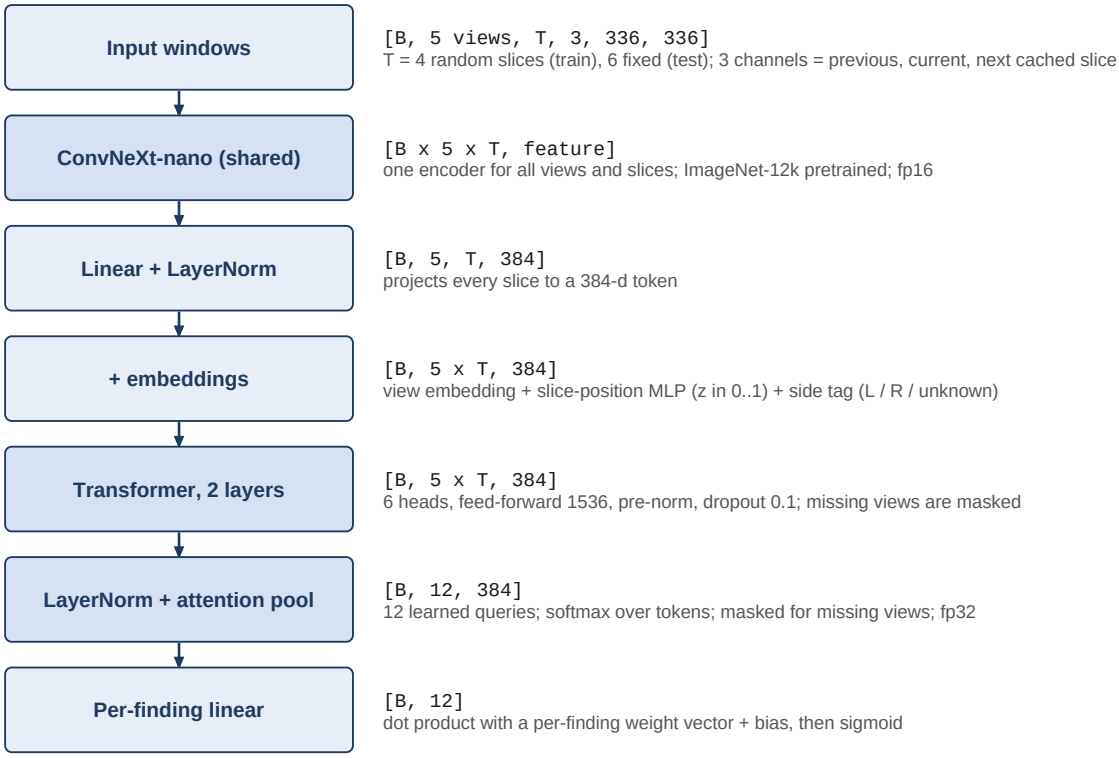


Figure 2. Data flow of one member. B is the batch size; T is the number of slices per view.

5. Training

5.1 Data and validation protocol

Each member trains on **all 4,349 weak studies** and is validated on the 58 gold studies only. A held-out weak fold would cost 25 % of the data for little information, since the gold set is the only trustworthy check. The saved weights are the exponential moving average (EMA) of the weights at the **last epoch**; no epoch is chosen by gold score, because 58 studies are too noisy to select on.

5.2 Settings

Setting	Value	Note
Epochs	24	v11 used 18; its gold AUC was still rising
Batch	4 studies x 4 accumulation steps = 16	gradient clipping at norm 2.0
Optimizer	AdamW	learning rate 1.5e-4 (encoder), 5e-4 (head); weight decay 0.05, none on biases and norms
Schedule	1 warm-up epoch, then cosine decay	if the time budget cannot fit all epochs, the schedule shortens itself; in this run epochs = planned = 24
EMA	decay 0.997	with the usual warm-up of the decay at the start
Stochastic depth	0.05	encoder drop-path rate
Windows per step	20 random (4 per view)	slices per view drawn from 4 jittered bins over the 12 cached slices
Loss	BCE with logits, soft targets	per-finding positive weight clip((1-p)/p, 1, 5), p = training prevalence
Seeds	142 and 143	42 + SEED0 (100) + member index; v11 used 42 and 43
Hardware	2 x T4, one member per GPU	both members train at the same time, no GPU-to-GPU communication

5.3 Augmentation

Done on the GPU, with the same random transform applied to all slices of one view so the 3-channel triplets stay consistent: rotation up to 15 degrees, scale of about 12.5 % either way, translation up to about 4 % of the image width, random gamma and a small brightness and contrast change. **No flips**, because left versus right carries meaning for several findings.

5.4 Loss details

The loss is binary cross-entropy against the fused soft labels. Findings with a low prevalence get a larger positive weight, capped at 5, so that the model is not pushed to predict "absent" for everything. A label mask exists for studies whose report could not be parsed (used only when the regex labeler is the only source); the loss is normalised by the number of valid label cells. Two ideas from earlier versions are **switched off** in v11 and v12: mixing out-of-fold predictions into the targets (distillation) and the pseudo-label memory (refinement). Public A/B tests in this competition reported that they lowered the leaderboard score.

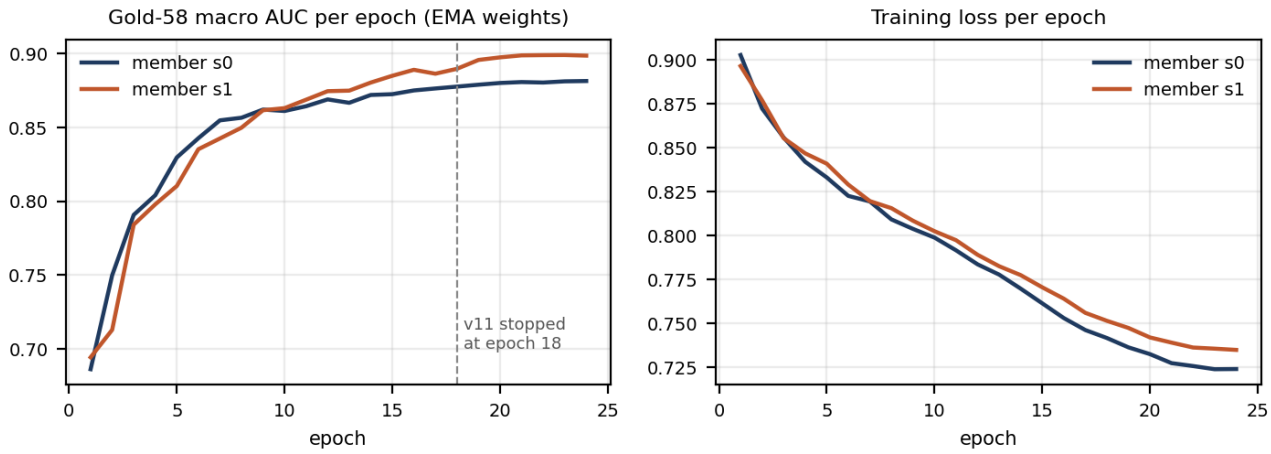


Figure 3. Training curves of the two v12 members (from history_v12_s0.csv and history_v12_s1.csv). Gold AUC is macro AUC over findings on the 58 gold studies; each epoch took about 10 to 11 minutes.

The curves show that the 24-epoch run reached a plateau: from epoch 21 to 24 the gold AUC moved by +0.0007 for member s0 and by -0.0002 for member s1. Between epoch 18 (where v11 stopped) and epoch 24 it rose from 0.878 to 0.881 (s0) and from 0.890 to 0.899 (s1). Both changes are well inside the gold-set standard error of about 0.02, so they are consistent with the longer schedule helping slightly but do not prove it.

6. Inference and ensembling

The submission cell reads the test studies, preprocesses them with the same function as the cache, and predicts with every model it finds.

- **Members.** All .pt files of the current version (2) and of the versions named in *KNEE_ENSEMBLE_WITH* (default *knee_v11*, 2 files) are loaded. The loader checks that their preprocessing settings match; v7 weights are refused because they used different pixels.
- **Averaging.** Sigmoid outputs are averaged over the 4 members and over 2 TTA passes. The two passes use two interleaved slice sets (offsets 0.25 and 0.75 inside each slice bin), so one study costs 4 models x 2 passes x 30 windows = 240 encoder forward passes.
- **Speed.** DICOM decoding runs in DataLoader worker processes while the GPUs process the previous batch. TTA switches itself off if the projected runtime would exceed 85 % of the inference budget (8 h of the 9 h Kaggle allows).
- **Fail-safes.** A series that fails to decode is masked; a study that fails completely gets 0.5 for every finding. A complete submission file is written every 100 studies, so a timeout still leaves a scoreable file. The final file is validated against sample_submission.csv: same studies, same column order, no NaN, values in [0, 1].
- **Optional blend with a public stack.** If a public stack's submission is present, both outputs are converted to per-finding percentile ranks and blended. The weight (grid 0 to 0.3) is chosen on the gold studies and stays 0 unless it gains at least 0.003 AUC; a weight of 0 skips this model's inference. This model also gives up and leaves the stack alone if it cannot finish before a hard limit of 8.5 h.

Why more members help

Averaging helps only if members make different mistakes. The two v12 members, which differ only in their seed, have a mean rank correlation of 0.93 on the gold studies (range 0.90 to 0.98 across findings), so there is real variation to average out. The notebook reports 0.85 to 0.95 per finding for the two v11 members.

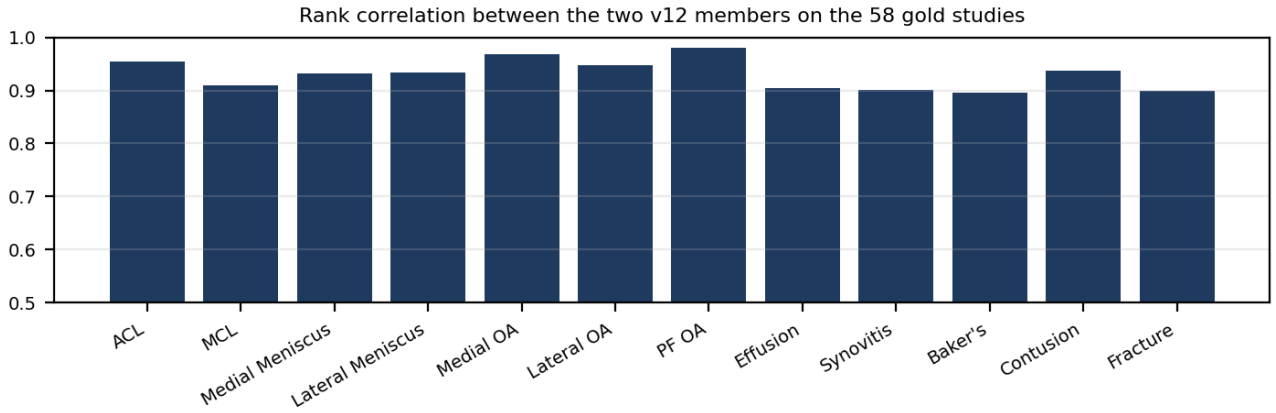


Figure 4. Spearman rank correlation of the two v12 members' predictions on the 58 gold studies, per finding (from oof_v12_s0.csv and oof_v12_s1.csv).

7. How the versions evolved

Version	Main idea	Score
v6	ConvNeXt-nano, 224 px, 4-fold training with a held-out weak fold	0.881 (leaderboard)
v7	ConvNeXt-tiny, 256 px, all weak data, leak-guarded label fusion, gated stack blend	0.897
v8	EfficientNet-B0, 320 px, 4 views	0.860 (worse than v7; not used)
v11	Only levers with public A/B evidence: longer training, 336 px, central 15-85 %, 20 random windows, ConvNeXt-nano, positive weights, no mirroring, no pseudo-labels	0.908
v12	v11 recipe + 2 more members (4 in total) + 24 epochs, fp32 weights, per-member seeding under a lock	members 0.881 / 0.899 on gold-58; hidden set not recorded

Table 3. Version history as stated in the notebook. Scores for v6 to v11 are leaderboard or hidden-set scores; the v12 numbers are gold-58 AUCs and are not comparable with them.

What v11 took from the public A/B study

v11 was built from the findings of a pre-registered A/B study by another team in the same competition (public repository TranBaDat2607/RSNA-Knee-Abnormality-Detection, docs/phase2_notes.md), which the notebook cites. The reported effects below are theirs, not measured by this project:

Lever	Their finding	v11 / v12 setting
Training length	20 epochs beat 12 (+0.0045 val); every run still improving at its end	18 in v11, 24 in v12
Resolution	336 px beat 256 px (+0.0025 val)	336 px at the same 153.6 mm field of view
Slice span	4-96 % lost 0.0055 against 15-85 %	15-85 %
Windows per step	16 random windows beat 40, and are 2.4 x cheaper	20 random windows (4 per view)
Backbone	ConvNeXt-tiny, ResNet-50 and CoAtNet-1 no better than nano	ConvNeXt-nano
Positive weighting	used in their BCE	clip((1-p)/p, 1, 5)
Mirroring left knees	clearly negative (-0.10 macro after 3 epochs)	off; side tag as an input
Mixing OOF predictions into targets	hurt: leaderboard 0.922 to 0.917 as more was mixed	off

The same study found that the remaining image-side choices change the score by only about 0.005 and that models agree at rank correlation of about 0.97, i.e. they are label-bound. Their pipeline differs from this one (public 44-slice corpus, MIL attention head, no study transformer), so the lift may not transfer in full.

8. Results and limits

Run	Measure	Value
v7	hidden set	0.897
v11	hidden set	0.908
v12 member s0	gold-58 macro AUC, final epoch	0.881 (239 min)
v12 member s1	gold-58 macro AUC, final epoch	0.899 (263 min)
v12 ensemble of 4	hidden set	not recorded

Table 4. Measured results. Use 0.908 (v11) as the reference when the v12 hidden score becomes available.

- **Statistical power.** 58 gold studies give a standard error of about 0.02 on the AUC. Gold AUC can show a large failure but cannot rank two close recipes. The hidden-set score is the real test.
- **Label-bound.** The fused weak labels themselves have a macro AUC of 0.891 against the gold studies (Table 2), and the members score 0.881 and 0.899. The model cannot be much better than the labels it learns from, which

is why further image-side changes are expected to move the score by small amounts only.

- **Optimism in the gold numbers.** The label-fusion weights were fit on the same 58 studies, so gold-based numbers are mildly optimistic.
- **Cost.** Inference runs 4 models with 2 passes, so the GPU part of the hidden run is roughly twice that of a 2-member ensemble. Setting *KNEE_ENSEMBLE_WITH=""* keeps only the v12 members.
- **Compatibility.** v7 weights and the v7 cache cannot be reused (different pixels). The v12 members and v11 members share preprocessing and can be mixed.

9. Reproducing the run

The notebook is written for Kaggle. Two runs are needed:

Run	MODE	Settings	Attach as input
1. Train	full	GPU T4 x 2, Internet ON, Save & Run All	competition data, the two label datasets, the v11 cache output (same cache tag, so no re-caching), the v11 pack
2. Submit	infer	GPU, Internet OFF	competition data, the v12 output, the v11 pack

Checks after run 1: no lines starting with !!, the number of epochs equals the planned 24, and the line "gold predictions of:" lists both v11 and v12 files. Check after run 2: the log says "ensemble of 4".

Environment variable	Default	Effect
`KNEE_ENSEMBLE_WITH`	knee_v11	packs whose members join the ensemble; empty string = own members only
`KNEE_MEMBERS`	2	number of members trained (one per GPU)
`KNEE_SEED0`	100	member seeds are 42 + SEED0 + index
`KNEE_EPOCHS`	24	training epochs
`KNEE_INPUT_ROOT`	/kaggle/input	where attached data is looked up
`KNEE_WORK`	/kaggle/working	where caches, weights and the submission are written
`KNEE_CANON`	0	1 = mirror left knees to one orientation (not recommended)
`KNEE_POSW`	5.0	cap of the positive weight in the loss
`KNEE_BUDGET_H`	9.0	training time budget in hours
`KNEE_HARD_LIMIT_H`	8.5	hard time limit for this model when blending with a stack

The trained weights (knee_v12_s0.pt, knee_v12_s1.pt) and the competition data are not part of the repository.

Glossary

Term	Meaning
Gold / weak labels	Gold: expert labels (58 studies). Weak: labels extracted automatically from radiology reports (4,349 studies).
AUC	Probability that a random positive study is scored above a random negative one; 0.5 is chance, 1.0 is perfect.
EMA	Exponential moving average of the weights during training; usually more stable than the last raw weights.
TTA	Test-time augmentation: predicting the same study twice with two different slice sets and averaging.
2.5D	A 2D network that sees a few neighbouring slices as channels, instead of a full 3D network.
OOF	Out-of-fold predictions on studies a model did not train on.